

Unit 03 Practice Problems: Probability

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Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and preparing for exams, make sure you have completed and reviewed all practice problems from the unit.

3.1

A basket contains a red ball, a green ball, and a yellow ball. One ball will be drawn out of the basket at random. If the ball is red or green, the experiment ends. If the ball is yellow, then another ball will be drawn out of the basket at random (the yellow ball is not placed back into the basket). On the second draw, if the ball is red then the experiment ends. If the ball is green on the second draw, then a fair coin is flipped once. Draw a tree diagram and identify all the outcomes of the Sample Space and their probabilities.

3.2

Redmond Elementary School has 3rd, 4th, and 5th grade classes. First, a class will be randomly selected and we will record if it is a class of 3rd, 4th, or 5th graders. Then, a student within the class will be selected at random and we will record if the student is a boy or girl. For example, one possible outcome in the Sample Space is 4G (4th grade class is selected, then a girl).

(a) Draw a tree diagram, then write out the Sample Space using set notation.

(b) Write the following events in set notation. For example: the event a third grader is selected = $\{3B, 3G\}$.

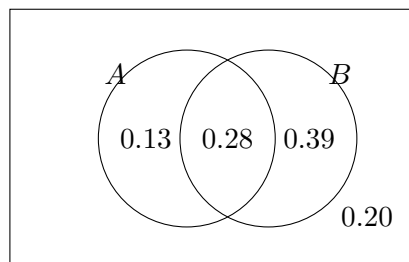
A = the selected student is a girl

B = the selected student is a 3rd grader or a boy

C = the selected student is not in 4th grade

3.3

Use this Venn Diagram to find the following probabilities. Also, shade the event on a picture of the diagram.



$$\begin{aligned}
 &P(A) \\
 &P(B) \\
 &P(A \cap B) \\
 &P(A \cup B) \\
 &P(A \cap \bar{B}) \\
 &P(\bar{A} \cap B) \\
 &P(\bar{A}) \\
 &P(\bar{B}) \\
 &P(\overline{A \cup B \cup (A \cap B)}) \\
 &P((A \cap \bar{B}) \cup (\bar{A} \cap B)) \\
 &P(A \cup \bar{B}) \\
 &P(\bar{A} \cup B) \\
 &P(\overline{A \cup B}) \\
 &P(\overline{A \cap B})
 \end{aligned}$$

3.4

If a male resident of a medical care facility is randomly selected, the probability he is over 80 years old = $P(A) = 0.50$. The probability he has prostate cancer = $P(B) = 0.32$. The probability he is both over 80 years old and has prostate cancer is 0.22.

(a) Draw a Venn Diagram with the given information.

(b) For each event below, describe the man who is randomly selected if the event occurs.

$$\begin{aligned}
 &A \cup B \\
 &\bar{A} \cap B \\
 &A \cup \bar{B}
 \end{aligned}$$

(c) Find the following probabilities.

$$\begin{aligned}
 &P(A \cup B) \\
 &P(\bar{A} \cap B) \\
 &P(A \cup \bar{B}) \\
 &P(\bar{A} \cap \bar{B}) \\
 &P(\bar{A} \cup \bar{B})
 \end{aligned}$$

3.5

The instructor of a statistics course holds an optional review session at 8 am. If a student enrolled in the class is selected at random, the probability they attend the review session = $P(R) = 0.60$. The probability the student attends the review session and has not drank any coffee that morning = $P(R \cap \bar{C}) = 0.06$. The probability the student does not attend the review session and also has not drank any coffee = $P(\bar{R} \cap \bar{C}) = 0.04$.

(a) Draw a Venn Diagram with the given information.

(b) Find the following probabilities and describe the student selected if the event occurs.

$$\begin{aligned}
 &P(\bar{C}) \\
 &P(R \cap \bar{R}) \\
 &P(C \cup \bar{C}) \\
 &P(\bar{R} \cap C)
 \end{aligned}$$

$$P(\overline{R} \cap \overline{C})$$

$$P(\overline{R \cup C} \cup (R \cap C))$$

$$P(\overline{R} \cup C)$$

3.6

Fred randomly selects one ball at random out of a basket. The balls in his basket are blue and numbered 7, 8, and 9. Gina also selects one ball out of a basket. The balls in her basket are red and numbered 5, 6, 7, 8, and 9. Fred and Gina will add the two numbers that appear.

(a) Draw a table and find the following probabilities:

$$P(\text{sum} = 13)$$

$$P(\text{sum} = 16)$$

$$P(\text{sum} > 14)$$

$$P(\text{sum} \leq 15 \cap \text{red} \leq 7)$$

$$P(\text{sum} \leq 15 \cup \text{red} \leq 7)$$

$$P(\text{sum} = 14 \cap \text{red} = \text{blue})$$

$$P(\text{sum} = 14 \cup \text{red} = \text{blue})$$

(b) What event(s) listed below are mutually exclusive with the event $\text{red} = \text{blue}$?

sum = 12
sum = 14
sum = 15
red < 7
red < 8
blue = 9

(c) What event(s) listed below are mutually exclusive with the event $\text{sum} = 13$?

sum = 17
blue = 9
red = 6
red = 7
sum < 16
blue = 7

3.7

Refer to Practice Problem 3.6. Find the following conditional probabilities:

$$P(\text{sum} = 16 \mid \text{red} = 8)$$

$$P(\text{blue} \geq 8 \mid \text{sum} = 16)$$

$$P(\text{sum} = 16 \mid \text{blue} \geq 8)$$

$$P(\text{sum} = 17 \mid \text{red} = 6)$$

$$P(\text{red} \leq 7 \mid \text{sum} = 13)$$

$$P(\text{blue} = 7 \mid \text{sum is even})$$

3.8

Refer to Practice Problem 3.1.

- (a) Given that the green ball appears during the experiment, what is the probability the yellow ball appears during the experiment?
- (b) Given that the experiment ends after the first draw, what is the probability the green ball appears during the experiment?
- (c) Are the events $A = \text{the first draw is yellow}$ and $B = \text{heads appears during the experiment}$ independent? Prove your answer.

3.9

Refer to Practice Problem 3.5.

- (a) Given that a randomly selected student attends the review session, what is the probability they drank coffee? Compare $P(C | R)$ to $P(C)$. What do you notice? What does this tell you about the events C and R ?
- (b) Based on your answer to the previous question, you should know in advance whether or not $P(C \cap R)$ equals $P(C) \cdot P(R)$. You should also know if $P(R | C)$ equals $P(R)$. Confirm your answers.
- (c) In terms of the story problem, describe in words what it means for C and R to be independent.

3.10

An anthropologist is studying a population of natives in Papua New Guinea, all residents of an East Tribe or a West Tribe. If a native is selected at random, the probability they belong to the East Tribe is .60 and the probability they belong to the West Tribe is .40. If a member of the East Tribe is selected at random, the probability they have blood type O is .50, blood type A is .35, blood type B is .10, and blood type AB is .05. If a member of the West Tribe is selected at random, the probability they have blood type O is .35, blood type A is .20, blood type B is .25, and blood type AB is .20.

- (a) Draw a tree diagram with probabilities written on the branches. At the end of each branch, identify each outcome of the Sample Space and its probability.
- (b) What is the probability a randomly selected native has blood type A?
- (c) Given that a randomly selected native has blood type AB, what is the probability they belong to the West Tribe?
- (d) Given that a randomly selected native does not have blood type O, what is the probability they belong to the East Tribe?

3.11

A company makes fluorescent light bulbs in three factories: Rockford, Sycamore, and Thompson. One bulb is selected at random.

70% of the bulbs are made in Rockford: $P(R) = .70$

20% of the bulbs are made in Sycamore: $P(S) = .20$

10% of the bulbs are made in Thompson: $P(T) = .10$

W = a bulb is still working after 10,000 hours of use

Rockford has a 90% success rate: $P(W | R) = .90$

Sycamore has a 80% success rate: $P(W | S) = .80$

Thompson has a 70% success rate: $P(W | T) = .70$

(a) Draw a tree diagram with probabilities written on the branches. At the end of each branch, identify each outcome of the Sample Space and its probability.

(b) What is the probability a randomly selected bulb is still working after 10,000 hours?

(c) Given that a randomly selected bulb is not working after 10,000 hours, what is the probability it was made in Thompson?

(d) Given that a randomly selected bulb is working after 10,000 hours, what is the probability it was made in Rockford or Sycamore?

(e) If the Rockford factory has the highest success rate, why does it produce the most defective bulbs? Doesn't that seem weird? Explain.

3.12

An unfair coin has $P(H) = .68$.

(a) If the unfair coin is flipped 5 times, what is the probability of observing HTTHT?

(b) If the unfair coin is flipped 7 times, what is the probability of observing tails at least once?

(c) If the unfair coin is flipped 8 times, what is the probability of observing five heads and three tails in any order? **Hint:** the number of ways to arrange 5 H's and 3 T's equals 56.

3.13

The integers on an unfair six-sided die appear with the following probabilities:

$P(\text{one})$	$P(\text{two})$	$P(\text{three})$	$P(\text{four})$	$P(\text{five})$	$P(\text{six})$
.17	.27	.07	.24	.09	.16

(a) If you roll the unfair die 4 times, what is the probability you will observe an even number on all 4 rolls?

(b) If you roll the unfair die 5 times, what is the probability you will observe a five at least once?

(c) If you roll the unfair die repeatedly, what is the probability you will observe a six for the first time on the 5th roll?

(d) If you roll the unfair die 3 times, what is the probability you will observe a two on exactly 1 of the 3 rolls? **Hint:** notice this question does not require the two to appear on the first, second, or third roll.

3.14

Eight friends each write their name on a card and place it in a hat. Ashley, Bob, Carla, Danny, and Edith write their names on yellow cards. Frank, Greg, and Hannah write their names on pink cards.

- (a) If three cards will be drawn one at a time at random with replacement, what is the probability all three cards will be pink?
- (b) Find the answer to the previous question if the cards are drawn without replacement.
- (c) If four cards will be drawn one at a time at random with replacement, what is the probability Hannah's name will appear zero times?
- (d) Find the answer to the previous question if the cards are drawn without replacement.
- (e) If two cards will be drawn one at a time at random with replacement, what is the probability they will be different colors?
- (f) Find the answer to the previous question if the cards are drawn without replacement.
- (g) If four cards will be drawn one at a time at random with replacement, what is the probability that at least one person's name will appear more than once?
- (h) Find the answer to the previous question if the cards are drawn without replacement.

3.15

Six high school students will choose a novel from the library to take home and read. Obviously, all the students must choose a different novel. The library has 894 novels. How many different arrangements are possible?

3.16

Twenty-five members of a hockey team must choose an integer between 1 and 99 to wear on the back of their hockey jersey. All the players must choose a different integer. How many different arrangements are possible?

3.17

21 balls are in a basket: 10 are red, 6 are green, 3 are yellow, and 2 are orange. Exactly one of the red balls has a happy face drawn on it. It is known as the "Red Super Happy Ball". Four balls will be drawn out of the basket all at once at random without replacement. Find the probabilities of the following events:

- (a) 2 balls are red, 1 is green, and 1 is orange.
- (b) All 4 balls are red.
- (c) None of the balls are green.
- (d) All of the balls are either red or green.
- (e) At least one of the balls is yellow.
- (f) 3 of the balls are red, and one of them is the Red Super Happy Ball.

(g) Two of the balls are either red or orange, the other two balls are either green or yellow, and the Red Super Happy Ball does not appear.

(h) There are zero yellow balls or there are zero green balls.

3.18

Fact:

$$\binom{15}{7} \binom{1}{1} + \binom{15}{8} \binom{1}{0} = \binom{16}{8}$$

(a) Write a very short story explaining why it should make sense that

$$\binom{15}{7} \binom{1}{1} + \binom{15}{8} \binom{1}{0} = \binom{16}{8}.$$

Your example could be about balls in baskets, people serving on committees, or whatever you like.

(b) Optional algebra challenge: prove the equality

$$\binom{n-1}{r-1} \binom{1}{1} + \binom{n-1}{r} \binom{1}{0} = \binom{n}{r}.$$

3.19

Draw a Venn Diagram with the given information.

$$P(A \cap B \cap C) = .07$$

$$P(A \cap B) = .09$$

$$P(A \cap C) = .17$$

$$P(B \cap C) = .10$$

$$P(A) = .41$$

$$P(B) = .20$$

$$P(C) = .36$$

(a) Find the following probabilities:

$$P(A \cap \overline{B} \cap \overline{C})$$

$$P(A \cup B \cup C)$$

$$P(A \cup C)$$

$$P((A \cup C) \cap B)$$

$$P(C \cup (A \cap B))$$

$$P((\overline{B} \cap C) \cap A)$$

(b) Are $A \cup C$ and B independent events? Prove your answer.

3.20

Solve the following exercises from the “Chapter 2 Exercises” section of the textbook (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger).

- 2.1.13, 2.2.2, 2.2.3, 2.2.7, 2.3.1, 2.3.7, 2.3.8, 2.4.3, 2.4.7, 2.6.1, and 2.7.1.

[To find these exercises in the digital copy of the textbook (**Textbooks > Files > Canvas**), go to the xvi Contents page and click the “EXERCISES” tab located at the lower-right corner of the page. This will open the complete list of chapter exercises.]

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.

Chapter 2 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 2.1.19, 2.1.23, 2.1.25, 2.2.4-2.2.6, 2.3.3, 2.3.17, 2.4.1, 2.5.1, 2.5.3, 2.6.3, 2.6.5, 2.6.11, 2.7.3, 2.7.7, 2.8.1, 2.8.2, 2.8.7, and 2.8.9.